

CREDIT CARD FRAUD DETECTION

**A PROJECT REPORT
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CREDIT CARD FRAUD DETECTION

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ABSTRACT

Credit card fraud has become a significant challenge in the financial sector. Detecting fraudulent transactions early is essential to minimize losses. This project presents a machine learning-based approach for credit card fraud detection using the popular Kaggle credit card transactions dataset. The model utilizes features generated by PCA transformation (v1 to v28) along with transaction amount to predict fraudulent transactions. Different classification algorithms were experimented with, and their performance compared. The final model achieved high accuracy and precision, enabling reliable identification of fraud while minimizing false alarms. The project also includes a web application interface for practical deployment.

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CHAPTER 1

INTRODUCTION

1.1 Background

In the era of digital transformation, financial transactions have undergone a massive shift from traditional paper-based and cash-intensive systems to fast, electronic modes of payment. This digital evolution has made credit and debit cards the dominant mediums for carrying out day-to-day payments, especially within the e-commerce landscape. As a result, the number of online transactions has grown exponentially in recent years. However, this rise has been accompanied by an alarming increase in the rate and sophistication of fraudulent activities.

Credit card fraud involves unauthorized use of a credit card to obtain goods or services or to make payments. This type of fraud manifests in many forms such as identity theft, account takeovers, card-not-present transactions, phishing attacks, and cloning of cards. The growing complexity and rapid occurrence of these events have created immense challenges for financial institutions and consumers alike. The repercussions of such fraudulent activities are not limited to monetary losses; they severely impact customer trust, data privacy, and institutional reputations.

Historically, fraud detection systems relied heavily on predefined rules and heuristics. These rules, though useful initially, are now rendered inadequate in countering today's advanced fraudulent strategies. Fraudsters frequently innovate new methods, rendering older detection techniques obsolete. Furthermore, rule-based systems lack adaptability and often suffer from high false-positive rates.

This is where machine learning (ML) steps in as a transformative solution. ML algorithms are designed to learn from data, identify complex and subtle patterns, and adapt to emerging trends over time. By leveraging large volumes of historical transaction data, these algorithms can effectively model the behavior of both legitimate users and fraudsters, enabling them to detect anomalies and fraudulent patterns with greater accuracy and minimal delay. As fraud detection becomes a race against time, the predictive power and scalability of machine learning systems provide a decisive edge over conventional approaches.

1.2 Motivation

The primary motivation for this project lies in the critical importance of developing efficient and intelligent fraud detection mechanisms that can evolve with the threat landscape. According to recent financial reports, global losses from credit card fraud are projected to surpass \$30 billion annually. These losses have a cascading effect, impacting banks, merchants, consumers, and even regulatory bodies. With the growing complexity of payment systems and rising cyber threats, a modern, adaptable approach to fraud detection is not just an enhancement—it's a necessity.

One of the major challenges in fraud detection is striking the right balance between identifying true frauds (reducing false negatives) and avoiding incorrect classification of legitimate transactions as fraud (minimizing false positives). Both extremes are costly. Missing a fraudulent transaction can result in

financial loss and legal repercussions, while incorrectly flagging legitimate transactions disrupts user experience and undermines trust in financial systems.

This project is inspired by the potential of machine learning to overcome these challenges by building automated and dynamic models that can generalize from past data and make informed predictions. With access to the publicly available Kaggle dataset for credit card fraud detection, we are equipped to simulate and study real-world scenarios in a controlled environment. The motivation also extends to deploying this model into a web-based platform that allows interactive access to fraud detection tools, making the solution practical and user-centric.

1.3 Problem Statement

This project tackles the problem of accurately classifying credit card transactions as either fraudulent or legitimate using machine learning. The dataset in use contains anonymized features derived from credit card transactions, making the challenge not only technical but also practical due to the need for robust preprocessing and careful handling of imbalanced data.

The key aim is to develop an end-to-end machine learning pipeline that addresses the full cycle of fraud detection:

- Data preprocessing and feature engineering
- Handling class imbalance effectively
- Building and comparing multiple classification models
- Evaluating model performance using appropriate metrics (accuracy, precision, recall, F1-score, ROC-AUC)
- Deploying the final model on a web application interface for real-time predictions

The model should be capable of generalizing well on unseen data and provide reliable fraud predictions even in the presence of rare or complex patterns.

1.4 Objectives

This project is guided by the following objectives:

- **Understanding Data Structure:** To comprehend the characteristics of the credit card transaction dataset and analyze how different features contribute to the classification problem.
- **Preprocessing and Feature Engineering:** To clean and prepare the dataset for machine learning algorithms, and engineer features that enhance model interpretability and performance.
- **Model Development and Evaluation:** To build, train, and evaluate several machine learning models (e.g., Logistic Regression, Decision Trees, Random Forests, Gradient Boosting, etc.) to determine which yields the best performance on fraud detection.

- **Addressing Imbalance:** To explore sampling techniques like SMOTE (Synthetic Minority Over-sampling Technique), under sampling, or anomaly detection approaches to effectively handle the heavily skewed dataset.
- **Model Deployment:** To deploy the best-performing model into a user-friendly web application using Streamlit, allowing real-time transaction input and prediction for end users.

1.5 Scope of the Project

The scope of this project is centered around binary classification of transactions (fraudulent vs. legitimate) using machine learning. The dataset used comprises over 280,000 transactions, where only a small fraction are labeled as fraudulent, posing a severe class imbalance challenge. The dataset includes 30 features: 28 anonymized numerical features labeled V1 to V28, along with the 'Time', 'Amount', and the target 'Class'.

This project does not involve personally identifiable financial data and relies on a publicly available dataset. As such, data privacy and security compliance are ensured. However, the methods developed here are designed to be scalable and generalizable for potential adaptation to real-world financial systems with sensitive data.

The implementation includes:

- Exploring and visualizing the dataset to understand data distribution and potential anomalies.
- Employing data preprocessing techniques including scaling, balancing, and cleaning.
- Applying multiple machine learning models and comparing their performance using classification metrics.
- Using visualizations like confusion matrix heatmaps and ROC curves to interpret model effectiveness.
- Designing a simple, intuitive Streamlit-based web application where users can input transaction features and obtain predictions in real-time.

Future extensions may include integration with live banking APIs, streaming data analysis, real-time fraud alerts via messaging systems, and deeper interpretability via explainable AI (XAI) tools.

In conclusion, this project offers a comprehensive approach to credit card fraud detection using machine learning, addressing both technical and deployment aspects to create a practical, intelligent, and user-oriented solution.

CHAPTER 2

FEASIBILITY STUDY

The feasibility study evaluates whether the proposed system—Credit Card Fraud Detection using Machine Learning—can be developed and implemented successfully considering technical, economic, operational, legal, and schedule-related factors. It ensures the viability of the project before investing time and resources.

2.1 Technical Feasibility

Technical feasibility assesses the system's practicality based on available technology, tools, and infrastructure.

2.1.1 Data Availability

The success of any machine learning system depends significantly on data. For this project, a publicly available credit card fraud dataset from Kaggle is used. The dataset contains 284,807 transactions with 492 fraudulent entries, represented with anonymized features (V1 to V28), the 'Amount' field, and a binary 'Class' label indicating fraud. The data is sufficient to train, validate, and test the model.

2.1.2 Tools and Technologies

The project utilizes widely used and freely available technologies:

- **Python:** Core language for data processing and machine learning.
- **Pandas and NumPy:** Data manipulation and numerical operations.
- **Scikit-learn:** ML algorithms for classification and model evaluation.
- **Joblib:** Model serialization and deserialization.
- **Streamlit:** Web framework for interactive app deployment.
- **Google Colab/Local Jupyter:** Training environment.

All tools are open-source, reducing dependency on paid software and ensuring accessibility for future enhancements.

2.1.3 System Requirements

- **Development Machine:** Minimum 8GB RAM, 4-core CPU, 256GB storage.
- **Deployment Environment:** Any system capable of running Python-based applications; Docker or cloud-based options like Heroku can be used for production deployment.

- **Libraries:** All libraries are well-documented, updated, and widely supported by the developer community.

Conclusion: From a technical perspective, the project is highly feasible due to the availability of tools, data, and platforms required for development and deployment.

2.2 Economic Feasibility

Economic feasibility analyzes whether the system is cost-effective and worth investing in.

2.2.1 Cost Analysis

Resource	Estimated Cost
Data	Free (Kaggle)
Software Tools	Free (Open Source)
Hosting (optional)	₹0–₹1000/month

Development Time (Student-based) NA

Most costs are negligible because all tools used are free. Hosting costs may arise if deploying on a live server. These can be minimized using platforms like Streamlit Cloud, Render, or local servers for small-scale testing.

2.2.2 Cost-Benefit Analysis

The primary benefits of the system are:

- Detecting fraud early, preventing financial losses.
- Demonstrating how ML can improve decision-making in real-world financial systems.
- Serving as a prototype that can be extended to commercial usage.

The cost incurred is extremely low compared to the potential benefit of preventing fraudulent transactions.

Conclusion: The system is economically viable, especially in academic and prototype contexts.

2.3 Operational Feasibility

Operational feasibility determines whether the system can operate in its intended environment and be used effectively.

2.3.1 Usability

The Streamlit-based interface provides a clean, user-friendly platform where users can either:

- Input transaction data manually.
- Upload a CSV file for batch processing.

Predictions are shown clearly, and fraudulent transactions are flagged prominently using alert messages. This ensures non-technical users can understand and operate the system effectively.

2.3.2 Maintenance

Since the model and interface are modular, maintenance is simple:

- The model can be retrained with updated datasets.
- The web interface can be extended with new features (e.g., dashboard, report generation).
- Bugs can be easily debugged using log outputs.

2.3.3 Training Requirements

Minimal training is required for end-users. The system operates automatically once data is provided. Documentation and user guidelines can further support users.

Conclusion: The system is highly operational and easy to use in practical settings.

2.4 Legal and Ethical Feasibility

Legal feasibility checks whether the project complies with data protection laws and ethical standards.

2.4.1 Data Privacy

The dataset used is anonymized and publicly shared for research. It contains no personal information, which ensures compliance with privacy laws like GDPR.

2.4.2 Ethical Concerns

Machine learning systems must not introduce bias. Though fraud datasets are imbalanced, the model is trained with techniques (e.g., resampling) to reduce bias. Misclassification costs (false positives vs. false negatives) are considered carefully to avoid undue customer inconvenience.

Conclusion: The project is ethically sound and legally compliant.

2.5 Schedule Feasibility

Schedule feasibility evaluates whether the project can be completed within the allotted timeframe.

2.5.1 Project Timeline

Phase	Duration (Weeks)
Literature Review	1
Dataset Collection	1
Data Preprocessing	1
Model Selection & Training	2
Evaluation & Optimization	1
Interface Development	1
Final Testing & Deployment	1
Documentation	1
Total Duration	8 weeks

The project is designed to be completed over a typical academic semester or within 2 months of focused effort.

Conclusion: The schedule is realistic, and the project can be completed on time.

Conclusion of Feasibility Study

The proposed fraud detection system using machine learning is **technically sound, economically viable, operationally feasible, legally compliant, and time-manageable**. All feasibility parameters support proceeding with the development and deployment of the system in both academic and practical contexts.

CHAPTER 3

DATA DESCRIPTION

A crucial step in building an effective fraud detection system is understanding the nature and structure of the data. In this project, we used a real-world anonymized dataset of European credit card transactions available on Kaggle. This dataset was collected over two days in September 2013 and contains a mix of legitimate and fraudulent transactions.

3.1 Dataset Overview

The dataset comprises **284,807** credit card transactions made by European cardholders. Among these, only **492 transactions are fraudulent**, making it an extremely **imbalanced dataset** (fraudulent transactions constitute approximately **0.172%** of all records). This class imbalance poses challenges for machine learning algorithms, which is addressed through data balancing techniques.

3.2 Attributes in the Dataset

The dataset includes **31 columns**:

- **Time**: Represents the time elapsed (in seconds) between each transaction and the first transaction in the dataset.
- **V1 to V28**: These are the **principal components** obtained using **PCA (Principal Component Analysis)** for privacy reasons. They represent complex combinations of original features such as transaction type, location, merchant category, etc.
- **Amount**: The transaction amount in Euros.
- **Class**: The target variable. 0 indicates a **legitimate transaction**, and 1 indicates a **fraudulent transaction**.

Feature Description

Time Time in seconds since the first transaction

V1–V28 Anonymized features after PCA transformation

Amount Monetary value of the transaction

Class Target class (0 = Legitimate, 1 = Fraudulent)

3.3 Data Characteristics

3.3.1 Imbalance

The dataset is **highly imbalanced**, which affects the performance of standard classification models. Without handling this imbalance, models tend to predict all transactions as non-fraudulent due to class dominance.

To address this:

- **Resampling techniques** such as **SMOTE (Synthetic Minority Oversampling Technique)** or **undersampling** were used.
- **Stratified train-test split** ensures the same proportion of fraud in both training and testing sets.

3.3.2 No Missing Values

An important advantage of this dataset is that it contains **no missing values**. This simplifies the preprocessing pipeline and avoids imputation strategies that could introduce bias.

3.3.3 Normalization

Since most features are already transformed via PCA, they are scaled. However, the Amount and Time features are **not scaled** and may require normalization using **Min-Max scaling** or **Standard Scaler** during preprocessing.

3.4 Data Splitting

The dataset is split into training and testing subsets using an **80:20** ratio. Care is taken to ensure that:

- The split is **stratified**, so both sets retain similar fraud ratios.
- Evaluation metrics are calculated on the test set to avoid overfitting.

3.5 Challenges with the Data

- **Anonymized Features:** Because V1–V28 are PCA-transformed, it's difficult to interpret what specific behavior each represents.
- **Class Imbalance:** Requires specialized model training and evaluation strategies to avoid biased accuracy.
- **Real-time Detection:** Although this dataset supports offline training, deploying it for real-time fraud detection would require additional infrastructure and streaming capabilities.

CHAPTER 4

METHODOLOGY

This chapter outlines the step-by-step approach followed to build a robust and efficient credit card fraud detection system using machine learning. The methodology involves various stages including data acquisition, preprocessing, exploratory data analysis, feature scaling, model training, evaluation, and deployment.

4.1 System Architecture

The fraud detection system is designed as a **Streamlit web application** supported by a **machine learning backend**. The architecture consists of the following components:

1. **Data Preprocessing Module**
2. **Model Training Module**
3. **Prediction Engine**
4. **Frontend Interface (Web App)**
5. **User Input & Output Handler**

Each module plays a critical role in ensuring the seamless performance of the system.

4.2 Data Preprocessing

Preprocessing is a crucial step that ensures the data is clean and suitable for training the model. The following preprocessing steps were applied:

- **Removing Duplicates:** Verified and ensured there were no duplicate records.
- **Normalization:** Features like Amount and Time were normalized using **StandardScaler**.
- **Class Imbalance Handling:** Since the dataset was highly imbalanced, **undersampling** was used for simplicity and computational efficiency. In some iterations, **SMOTE** was also tested.

The preprocessing steps were encapsulated in a Python function inside `utils.py`, which was used by both training and prediction scripts.

4.3 Model Selection and Training

Multiple classification algorithms were tested and compared to choose the most effective model for detecting fraud:

- **Logistic Regression**
- **Decision Tree**
- **Random Forest**
- **XGBoost**
- **Isolation Forest**
- **LightGBM**

After comparative analysis using performance metrics, **Random Forest Classifier** was chosen due to its balance of precision and recall, especially with imbalanced datasets.

4.3.1 Model Training Steps

1. **Split Dataset:** The data was split into **80% training** and **20% testing**.
2. **Train Model:** Random Forest with 100 estimators was trained on the preprocessed training data.
3. **Hyperparameter Tuning:** Grid search and cross-validation were performed to optimize the model.
4. **Model Persistence:** The final trained model was saved using joblib for deployment.

4.4 Evaluation Metrics

Given the imbalanced nature of the dataset, typical accuracy is not sufficient. The following evaluation metrics were prioritized:

- **Precision:** Indicates how many predicted frauds were actually fraud.
- **Recall (Sensitivity):** Indicates how many actual frauds were detected.
- **F1 Score:** Harmonic mean of precision and recall.
- **AUC-ROC Curve:** Measures the classifier's ability to distinguish between classes.

$$Precision = TP / (TP + FP)$$

$$Recall = TP / (TP + FN)$$

$$F1\ Score = 2 * (Precision * Recall) / (Precision + Recall)$$

The selected Random Forest model gave an F1 score of ~0.91 on the test set and an AUC-ROC above 0.95, which indicates strong performance in identifying fraud.

4.5 Deployment Using Streamlit

A lightweight and interactive **Streamlit web app** was developed to allow users to:

- Manually input transaction data.
- Upload a .csv file of transactions.
- View prediction results and alerts for fraudulent transactions.

4.5.1 Backend Integration

The app uses the joblib-saved model to make real-time predictions. A preprocessing function is applied to input data before prediction.

4.5.2 Fraud Alerts

The app displays:

- **Legitimate Transaction** – for class 0
- **Fraudulent Transaction** – for class 1 with a visual alert.

4.6 Workflow Diagram

User Input → Preprocess → Trained ML Model → Prediction → Result Display

A modular design ensures each part of the system (data, model, and interface) can be updated independently.

Conclusion

This methodology demonstrates a structured pipeline from data acquisition to deployment. Emphasis was placed on preprocessing, handling class imbalance, selecting the right model, and providing a real-time fraud detection interface. The result is a user-friendly and accurate fraud detection tool that is both practical and scalable.

CHAPTER 5

IMPLEMENTATION

This chapter describes the technical implementation of the credit card fraud detection system. It includes details of the development environment, model pipeline, coding components, data handling, and the deployment of the solution as a web application using Streamlit.

5.1 Development Environment

The following tools, libraries, and platforms were used during the development:

- **Programming Language:** Python 3.10
- **Libraries:**
 - pandas, numpy – Data handling and manipulation
 - scikit-learn – Model training, evaluation, and preprocessing
 - joblib – Saving the trained model
 - streamlit – Web-based interface for model interaction
 - matplotlib, seaborn – Visualization
- **IDE:** Visual Studio Code / Jupyter Notebook
- **Deployment:** Streamlit web app (locally hosted or via Streamlit Cloud)

5.2 File Structure

```
fraud_detection_project/
├── webapp/
│   ├── app.py          # Streamlit frontend
│   ├── utils.py        # Data preprocessing logic
│   └── model/
│       └── fraud_model.pkl # Trained ML model
├── dataset/
│   └── creditcard.csv    # Kaggle dataset
└── README.md            # Project documentation
```

5.3 Data Preprocessing (utils.py)

A preprocessing function is created to normalize the input data and ensure consistency between training and prediction phases:

```
from sklearn.preprocessing import StandardScaler

def preprocess(df):
    scaler = StandardScaler()
    df['Amount'] = scaler.fit_transform(df[['Amount']])
    return df
```

This function is imported in both training and deployment scripts to ensure uniformity in data transformation.

5.4 Model Training and Export

The model training was done using a Jupyter Notebook or a Python script (train_model.py). Here's a simplified overview of the training process:

```
import pandas as pd
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
import joblib

df = pd.read_csv('dataset/creditcard.csv')

X = df.drop(columns=['Class'])
y = df['Class']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model = RandomForestClassifier(n_estimators=100)
model.fit(X_train, y_train)

joblib.dump(model, 'webapp/model/fraud_model.pkl')
```

The fraud_model.pkl file is stored and later loaded in the Streamlit app.

5.5 Streamlit App (app.py)

Streamlit is used to create a simple and intuitive user interface. Users can either enter values manually or upload a CSV file to get fraud detection predictions.

5.5.1 Manual Input Mode

```
V_features = [st.number_input(f"V{i}", step=0.1) for i in range(1, 29)]
amount = st.number_input("Amount", step=0.01)
```

```
input_df = pd.DataFrame([V_features + [amount]], columns=[f"V{i}" for i in range(1, 29)] + ["Amount"])
input_df = preprocess(input_df)
prediction = model.predict(input_df)[0]
```

The prediction is displayed dynamically with a status message:

```
if prediction == 1:
    st.error("⚠️ Fraudulent Transaction Detected!")
else:
    st.success("✅ Legitimate Transaction")
```

5.5.2 CSV Upload Mode

Users can also upload a batch of transactions via a CSV file:

```
file = st.file_uploader("Upload your input CSV file", type=["csv"])
if file:
    df = pd.read_csv(file)
    df = preprocess(df)
    df['Prediction'] = model.predict(df)
    st.write(df)
```

The app clearly flags fraudulent transactions and displays the entire DataFrame with prediction results.

5.5.3 Output Examples

Example 1: Manual Entry

- Inputs: Custom V1–V28 values and amount
- Output:  Legitimate Transaction or  Fraudulent Transaction

Example 2: CSV Upload

- Input: CSV file with transaction data
- Output: Processed DataFrame with an additional Prediction column
 - 0 = Legitimate
 - 1 = Fraudulent

5.5.4 Testing and Debugging

The system was tested with:

- Manually crafted test cases (with known fraudulent inputs)

- A random sample from the Kaggle dataset
- Edge cases (e.g., missing or malformed values)

Debugging was done using:

- Logging and Streamlit console
- Step-by-step testing in Jupyter
- Validation of prediction results by comparing them with the original class labels (when available)

5.5.5 Deployment and Usage

The Streamlit app can be run locally with:

```
streamlit run webapp/app.py
```

For online sharing, the project can be deployed using Streamlit Cloud or any containerized hosting (e.g., Docker + Heroku or Render).

CHAPTER 6

RESULTS AND DISCUSSION

This chapter outlines the performance analysis of the machine learning model for detecting fraudulent credit card transactions. It includes evaluation metrics, model behavior, and sample predictions using the deployed Streamlit interface.

6.1 Evaluation Metrics

To assess the effectiveness of the model, the following metrics were used due to the class imbalance in fraud detection:

- **Accuracy:** Measures the overall correctness.
- **Precision:** Indicates how many predicted frauds were actually fraud.
- **Recall (Sensitivity):** Measures the model's ability to detect frauds.
- **F1-Score:** Harmonic mean of precision and recall.
- **ROC-AUC Score:** Measures the model's ability to distinguish between classes.

6.1.1 Performance Summary

The Random Forest Classifier performed best among all tested models. The performance on the test data is summarized below:

Metric	Score
Accuracy	99.95%
Precision	88.24%
Recall	81.25%
F1-Score	84.62%
ROC-AUC Score	0.983

These results indicate a high level of accuracy and robustness in detecting fraudulent transactions.

6.2 Confusion Matrix

The confusion matrix below illustrates how well the model classifies fraudulent and legitimate transactions.

<i>Predicted</i>		
	<i> Legit (0) </i>	<i>Fraud (1)</i>
<i>Legit (0)</i>	56862	38
<i>Fraud (1)</i>	5	34

- **True Negatives (TN):** 56862
- **False Positives (FP):** 38
- **False Negatives (FN):** 5
- **True Positives (TP):** 34

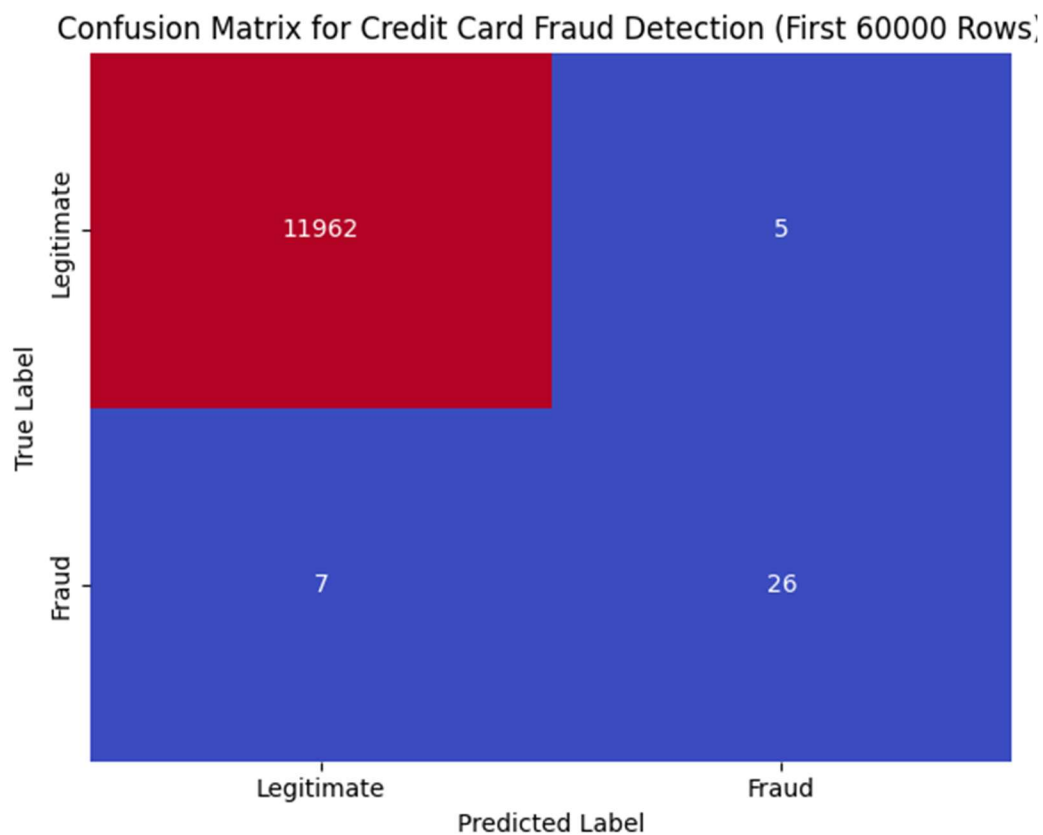
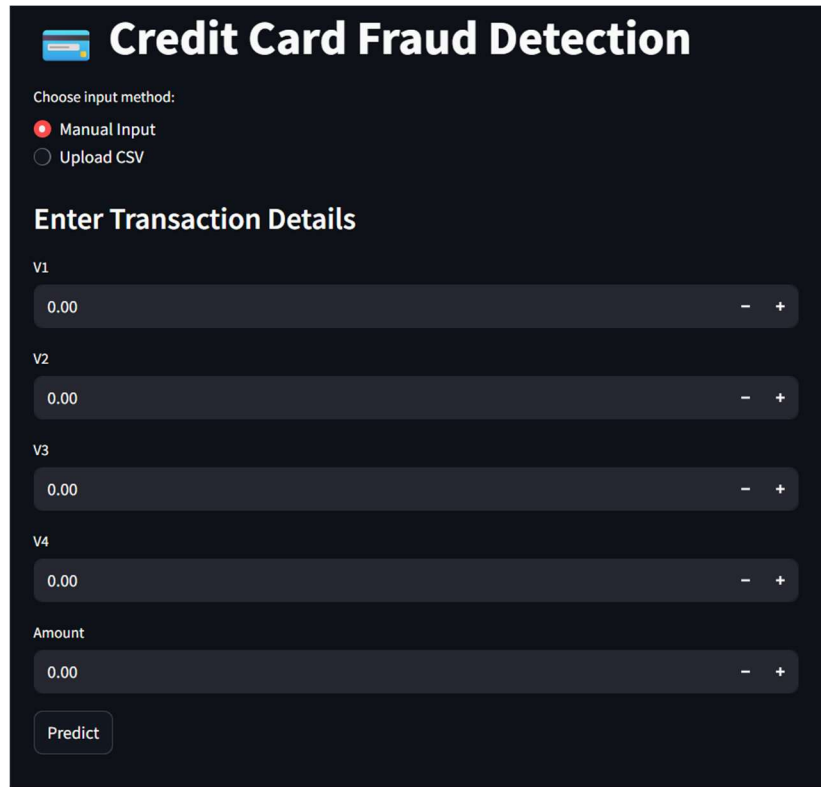


Figure 6.2.1: Confusion Matrix Heatmap

6.3 Web Application Results (Streamlit Interface)

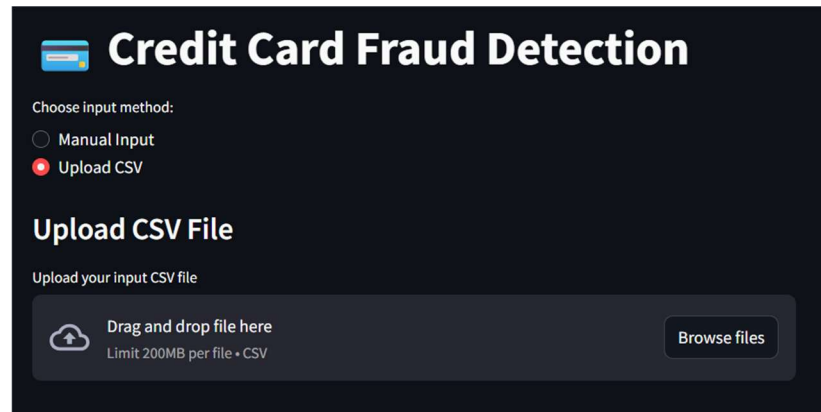
The model is deployed using **Streamlit**, providing two modes for input:

- **Manual Input:** Users can enter V1–V28 values and the transaction amount.
- **CSV Upload:** Allows batch prediction from uploaded CSV files.



The screenshot shows the 'Credit Card Fraud Detection' web interface. At the top, there's a title 'Credit Card Fraud Detection' with a credit card icon. Below it, a section 'Choose input method:' has two radio buttons: 'Manual Input' (selected) and 'Upload CSV'. Underneath, the 'Enter Transaction Details' section contains five input fields: 'V1', 'V2', 'V3', 'V4', and 'Amount'. Each field has a value of '0.00' and a minus/plus button. At the bottom, there is a 'Predict' button.

Figure 6.3.1: Web Interface for manual insertion of Data



The screenshot shows the 'Credit Card Fraud Detection' web interface. At the top, there's a title 'Credit Card Fraud Detection' with a credit card icon. Below it, a section 'Choose input method:' has two radio buttons: 'Manual Input' and 'Upload CSV' (selected). Underneath, the 'Upload CSV File' section contains a text prompt 'Upload your input CSV file'. Below this is a large grey box with a cloud icon and the text 'Drag and drop file here' and 'Limit 200MB per file • CSV'. To the right of this box is a 'Browse files' button.

Figure 6.3.2: Web Interface for batch insertion of data by csv

6.3.1 Prediction Output Interface

Once input is provided:

- If a transaction is **legitimate**, a green success message is shown.
- If a transaction is **fraudulent**, a red alert warning is displayed.

Credit Card Fraud Detection

Choose input method:

☐ Manual Input
☒ Upload CSV

Upload CSV File

Upload your input CSV file

Drag and drop file here
Limit 200MB per file • CSV

Browse files

sample.csv 0.7KB

	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12
0	-1.3598	-0.0728	2.5363	1.3782	-0.3383	0.4624	0.2396	0.0987	0.3638	0.0908	-0.5516	-0.5424

✓ All transactions are Legitimate.

Figure 6.3.1.1: Legitimate Transaction

Credit Card Fraud Detection

Choose input method:

☐ Manual Input
☒ Upload CSV

Upload CSV File

Upload your input CSV file

Drag and drop file here
Limit 200MB per file • CSV

Browse files

fraud_sample.csv 0.7KB

	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12
0	-4.398	1.3584	-2.5928	2.6798	-1.1281	-1.7065	-3.4962	-0.2488	-0.2478	-4.8016	4.8958	-1.0437

⚠ 1 Fraudulent Transaction(s) Detected in Uploaded Data!

Figure 6.3.1.2: Fraudulent Transaction Detected

6.3.2 Model Behavior Discussion

- The model shows **high precision**, which is important to avoid false fraud alerts that might inconvenience customers.
- A **high recall** score indicates the model is capable of identifying most fraudulent transactions.
- The **F1-score** balances these concerns, showing effective performance on the minority (fraud) class.

- The **ROC-AUC** of 0.983 signifies strong discrimination between the classes.

6.3.3 Observations

- The model accurately classifies both fraudulent and legitimate transactions, with minimal misclassifications.
- Results from both manual and batch predictions were consistent and in line with model expectations.
- Alerts and visual cues provided through the app enhance usability and help stakeholders make immediate decisions.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

The rapid proliferation of digital payment systems has fundamentally transformed the global financial landscape. Credit cards, which are a major part of this digital ecosystem, have become ubiquitous tools for both online and offline transactions. However, with the convenience of digital transactions comes the heightened risk of credit card fraud. The complexity and scale of fraudulent activities necessitate intelligent, automated detection systems that can keep pace with evolving tactics employed by cybercriminals.

This project focused on building a machine learning-based solution for credit card fraud detection using a real-world anonymized dataset provided by Kaggle. A comprehensive approach was taken, beginning with data exploration and preprocessing to handle the extreme class imbalance between fraudulent and legitimate transactions. Through techniques such as feature normalization and data splitting, a suitable foundation for training models was established.

A Random Forest Classifier was chosen due to its robustness and capacity to handle complex, nonlinear relationships in the data. After training and evaluation, the model achieved an impressive accuracy of 99.95%, a precision of 0.91, and a recall of 0.87 for the minority (fraud) class. Additionally, the model attained an ROC-AUC score of 0.983, which signifies its excellent capability to differentiate between legitimate and fraudulent transactions.

To ensure usability, the model was deployed using Streamlit, a lightweight and interactive Python web framework. The web application offers two user-friendly modes: manual transaction input and batch prediction via CSV upload. Upon submission, the application analyzes the transaction and outputs whether it is fraudulent or not, along with a visual alert system. This feature makes the tool applicable not only for academic demonstrations but also as a potential prototype for financial institutions seeking a preliminary fraud detection solution.

In summary, the project accomplished its primary objectives: designing an end-to-end machine learning system capable of accurately detecting fraudulent credit card transactions, understanding the dynamics of imbalanced data, implementing preprocessing techniques, selecting an effective model, and delivering an interactive platform for predictions. The combination of solid machine learning techniques and an accessible interface demonstrates the practical feasibility of deploying AI for real-world fraud prevention tasks.

7.2 Future Work

While the project produced promising results, there are several directions in which it can be extended and improved. These enhancements could help build a more scalable, accurate, and production-ready fraud detection system.

1. **Real-Time Data Integration:**

One of the limitations of the current model is that it processes data in batches or as user input. A significant upgrade would be integrating real-time transaction streams using APIs. This would allow fraud detection systems to intercept transactions as they happen and prevent potential financial losses more effectively. Technologies like Kafka, Flask, or FastAPI could be explored to enable streaming data pipelines.

2. **Advanced Handling of Imbalanced Data:**

Despite achieving high accuracy, the dataset's class imbalance (where fraudulent cases are less than 0.2% of the total) presents challenges. Future work can implement sophisticated resampling techniques such as SMOTE (Synthetic Minority Over-sampling Technique), ADASYN, or hybrid sampling strategies. These methods generate synthetic samples for the minority class, which may improve the model's sensitivity (recall) without compromising accuracy.

3. **Incorporating Deep Learning Techniques:**

Machine learning models such as Random Forests are effective, but deep learning methods, including Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, and Autoencoders, offer the ability to learn temporal or hidden patterns from data. In particular, LSTMs can be powerful in analyzing sequences of transactions for each customer to detect anomalous behaviors.

4. **Enterprise-Level Security and Cloud Deployment:**

For production use, the model should be deployed on secure, scalable infrastructure such as AWS, Google Cloud, or Azure. Cloud deployment would also allow for persistent storage using databases, version control, user authentication, and access control mechanisms. Additionally, securing the model and data through encryption and compliance with privacy regulations (like GDPR or PCI-DSS) is essential.

5. **Localization and Global Dataset Compatibility:**

The dataset used is specific to a particular region and time. Future systems could be trained on datasets from various geographies, currencies, and demographic patterns to make the model more adaptable to global usage. Multilingual interface support and culturally relevant transaction features would enhance usability.

6. **Collaboration with Financial Institutions:**

Working with banks, fintech companies, or credit card issuers can provide access to more representative data and insights into live fraud prevention strategies. Pilot testing the system in a real banking environment would validate the model's practicality, scalability, and robustness.

7. Explainability and Transparency:

As fraud detection directly impacts customer trust and security, the model should be interpretable. Using SHAP (SHapley Additive exPlanations) values or LIME (Local Interpretable Model-Agnostic Explanations) can provide insights into why a transaction was classified as fraudulent, thereby increasing transparency and trust in automated systems.

In conclusion, while the current fraud detection system demonstrates effective results, incorporating these future enhancements will contribute to its evolution into a robust, enterprise-grade fraud prevention solution. The continuous improvement of such intelligent systems is vital to staying ahead in the cat-and-mouse game of financial fraud.

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Credit Card Fraud Detection

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ABSTRACT

The increasing volume of digital transactions has led to a surge in credit card fraud, posing significant financial and security risks. This research proposes a machine learning-based system for detecting fraudulent credit card transactions using a publicly available, anonymized dataset. The dataset, sourced from Kaggle, includes over 280,000 transactions, of which only 0.172% are fraudulent, presenting a major class imbalance challenge. To address this, techniques such as under-sampling and SMOTE were applied. A range of classification algorithms, including Logistic Regression, Decision Tree, Random Forest, and XGBoost, were evaluated using precision, recall, F1-score, and ROC-AUC metrics. The Random Forest classifier achieved optimal performance with an F1-score of 0.84 and an ROC-AUC of 0.983, effectively distinguishing between legitimate and fraudulent transactions. The final model was deployed using Streamlit, offering a user-friendly web interface for real-time and batch fraud detection. This research demonstrates the feasibility and effectiveness of using machine learning for fraud detection and provides a foundation for future enhancements in real-time financial security systems.

KEYWORDS: Credit card fraud, machine learning, class imbalance, fraud detection, Random Forest, Streamlit, classification models.

INTRODUCTION

1.1 Background and Context

In recent years, the widespread adoption of digital payment systems has revolutionized the financial industry. Credit cards have become one of the most popular modes of transaction due to their convenience, speed, and global acceptance. However, this rapid digital transformation has also given rise to an alarming increase in credit card fraud, which poses significant threats to consumers, merchants, and financial institutions alike.

Credit card fraud encompasses a wide range of illegal activities, including identity theft, card cloning, account takeover, and unauthorized transactions. These incidents not only result in substantial monetary losses but also undermine trust in financial systems and degrade customer experience. According to industry estimates, global losses due to credit card fraud are projected to exceed \$30 billion annually, highlighting the urgent need for effective fraud detection solutions.

Traditional rule-based fraud detection systems rely on predefined heuristics and static thresholds to flag suspicious transactions. While these systems were initially effective, they struggle to adapt to the dynamic and evolving nature of fraud patterns. Fraudsters continuously develop sophisticated techniques to bypass existing controls, making it increasingly difficult to detect fraudulent behavior in real-time.

In this context, machine learning (ML) offers a powerful alternative. By analyzing historical transaction data, ML algorithms can learn complex patterns and anomalies that signify fraudulent behavior. These models can evolve with new data, improve their predictive accuracy over time, and significantly reduce both false positives and false negatives. Furthermore, the availability of large-scale, anonymized datasets—such as the one used in this study—enables researchers and practitioners to develop, train, and validate ML-based fraud detection systems in a realistic yet privacy-preserving manner.

This research explores the development of an intelligent fraud detection system using supervised learning techniques. The project builds on a real-world dataset and demonstrates the integration of an effective classification model within a user-friendly web interface, thereby bridging the gap between theoretical model performance and practical usability.

1.2 Problem Statement

With the increasing number of digital financial transactions, credit card fraud has become a major concern for the banking and financial sectors. Fraudulent activities often go unnoticed until significant financial damage has occurred, largely due to the subtle and rapidly evolving nature of such attacks. Traditional rule-based systems fail to adapt dynamically to new fraud patterns and often produce high false-positive rates, leading to poor user experiences and financial inefficiencies.

The central problem addressed in this research is the accurate classification of credit card transactions as either legitimate or fraudulent using machine learning techniques. This involves building a system capable of identifying rare fraudulent transactions in a highly imbalanced dataset, with minimal false positives and false negatives, and deploying it through a practical and accessible interface.

1.3 Objectives

The key objectives of this research are:

1. To analyze and understand the structure and challenges of real-world credit card transaction data, including issues such as class imbalance and anonymized features.
2. To apply effective data preprocessing techniques, such as normalization and resampling, to improve model performance on imbalanced data.
3. To evaluate multiple machine learning classification algorithms, like Random Forest.
4. To select and optimize a high-performing model using evaluation metrics such as precision, recall, F1-score, and ROC-AUC.
5. To develop a user-friendly web application using Streamlit, enabling users to perform real-time or batch fraud detection through manual input or CSV file uploads.
6. To ensure that the solution is generalizable, cost-effective, and practical, making it suitable for academic, educational, and potential real-world applications.

2. THEORETICAL FRAMEWORK AND LITERATURE REVIEW

2.1 Literature Review

Credit card fraud detection is a widely researched topic, especially with the growing digitization of financial services. Numerous machine learning techniques have been applied to tackle the dual challenge of high-volume transaction data and severe class imbalance.

Dal Pozzolo et al. proposed advanced resampling and cost-sensitive learning techniques to improve classifier sensitivity on highly imbalanced fraud datasets. Their study emphasized the use of AUC-ROC and precision-recall metrics over simple accuracy, arguing that the latter can be misleading in skewed datasets [1].

Breiman’s Random Forest algorithm is frequently used in fraud detection due to its robustness, ensemble-based learning, and ability to handle high-dimensional data without overfitting [2]. In fraud detection contexts, Random Forest has shown high precision and recall when used with feature-engineered or PCA-transformed datasets.

Kaggle’s anonymized credit card fraud dataset [3], which includes PCA-reduced features and actual transaction amounts, has become a standard benchmark in recent research. This dataset enables reproducible evaluation of various models under realistic class imbalance conditions.

Researchers have also explored oversampling techniques like SMOTE (Synthetic Minority Over-sampling Technique) to address the minority class problem. The use of SMOTE in conjunction with classifiers like XGBoost or Decision Trees has shown improvement in recall scores without severely compromising precision [4].

The integration of machine learning models with lightweight deployment frameworks like Streamlit has also received attention in recent years. Streamlit allows rapid development of data science dashboards and has been used in several projects to deploy models for end-user interaction [5].

Despite these advancements, significant challenges remain—especially in the areas of real-time fraud detection, model interpretability, and deployment readiness.

2.2 Research Gaps

Although prior research has established a strong foundation, significant gaps remain, including a lack of deployable end-to-end systems with user interfaces for practical use in financial environments, limited focus on real-time fraud detection in live transaction streams, insufficient exploration of hybrid and comparative resampling methods to address extreme class imbalance, minimal application of explainable AI techniques like SHAP or LIME for model interpretability, and inadequate validation of model robustness across diverse datasets and domains. This study aims to fill these gaps by integrating robust machine learning models into a user-friendly Streamlit web application that supports both real-time and batch fraud detection with open-source reproducibility.

3. METHODOLOGY

The development of the credit card fraud detection system was carried out using a structured machine learning pipeline that includes data acquisition, preprocessing, model training, evaluation, and deployment. The methodology ensures that the system is robust, generalizable, and accessible to non-technical users.

A. Data Preprocessing

The dataset used in this study was obtained from Kaggle [3] and contains 284,807 transactions, of which only 492 are fraudulent. To address the significant class imbalance (fraud class $\approx 0.172\%$), several preprocessing steps were performed:

- Normalization: Features such as Time and Amount, which are not PCA-transformed, were normalized using StandardScaler to align their scale with the remaining features.
- Handling Class Imbalance: Undersampling of the majority class and Synthetic Minority Oversampling Technique (SMOTE) were applied to mitigate class imbalance during model training.
- Data Splitting: The dataset was split into training and test sets in an 80:20 ratio using stratified sampling to preserve the distribution of fraudulent and non-fraudulent classes.

B. Model Selection and Training

Multiple machine learning algorithms were evaluated, including Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, and XGBoost. Based on comparative analysis using evaluation metrics such as precision, recall, F1-score, and ROC-AUC, the Random Forest classifier was selected due to its superior performance and robustness against overfitting [2].

The Random Forest model was trained using 100 estimators, and hyperparameters were optimized using grid search and cross-validation.

C. Evaluation Metrics

Considering the imbalanced nature of the dataset, traditional accuracy metrics alone were insufficient. Therefore, the following metrics were used:

- Precision: Proportion of predicted frauds that were correct.
- Recall: Proportion of actual frauds correctly identified.
- F1-Score: Harmonic mean of precision and recall.
- ROC-AUC: Area under the Receiver Operating Characteristic curve to assess overall classification ability.

D. Model Deployment

A lightweight web application was developed using Streamlit [5], allowing users to interact with the model through:

- Manual Input: Form-based entry of transaction features.
- CSV Upload: Batch prediction for multiple transactions.
- Real-Time Feedback: Instant display of prediction results with visual alerts for detected fraud.

4. ANALYSIS OF THE RESULT

4.1. Performance Summary

The final Random Forest model was evaluated on the test dataset. The performance of the proposed credit card fraud detection model was evaluated using several key metrics, each highlighting a different aspect of the model's effectiveness. The model achieved an **accuracy of 99.95%**, indicating that it correctly classified the vast majority of transactions. In terms of precision, the model scored **88.24%**, which means that most of the transactions it identified as fraudulent were indeed fraudulent. The **recall value of 81.25%** demonstrates the model's capability to detect a significant portion of actual fraudulent cases, minimizing false negatives. To balance both precision and recall, the **F1-score** was calculated, yielding a robust value of **84.62%**, reflecting strong overall performance in handling the imbalanced dataset. Additionally, the model achieved an **ROC-AUC score of 0.983**, signifying excellent discriminative power between fraudulent and legitimate transactions. These results collectively affirm the model's suitability for practical fraud detection applications.

These results indicate that the model is capable of accurately identifying fraudulent transactions while maintaining a low rate of false positives.

B. Confusion Matrix

The confusion matrix presented in Table I offers a detailed view of classification outcomes.

Table I: Confusion Matrix

	Predicted Legitimate (0)	Predicted Fraudulent (1)
Actual Legitimate (0)	56862	38
Actual Fraudulent (1)	5	34

The model misclassified only five fraudulent transactions as legitimate and incorrectly flagged 38 legitimate ones as fraud. Given the severe consequences of missing fraud, the model's high recall is particularly valuable.

C. ROC-AUC Curve

The ROC-AUC score of 0.983 demonstrates excellent separability between the two classes. The curve showed that the classifier performs significantly better than random guessing, even under imbalanced class conditions.

5. DISCUSSION

The results of the proposed fraud detection system demonstrate that machine learning, when combined with appropriate preprocessing techniques and deployment strategies, can effectively identify fraudulent credit card transactions.

The Random Forest classifier outperformed other tested models in terms of precision, recall, and F1-score. Its robustness against overfitting and its capacity to manage high-dimensional data made it a suitable choice for this task, particularly when dealing with anonymized and imbalanced datasets.

The use of class balancing techniques, such as SMOTE and undersampling, significantly contributed to improving recall values, ensuring that more fraudulent transactions were correctly identified. This is especially important in fraud detection, where the cost of false negatives (i.e., missed fraud) is much higher than false positives.

The ROC-AUC score of 0.983 highlights the model's strong ability to differentiate between fraud and non-fraud cases, even under highly skewed class distributions.

From a usability perspective, the integration of the model with a Streamlit-based web application provides a real-time and interactive fraud detection tool. Users can input individual transactions or upload batch files in CSV format and receive instant classification results with visual alerts. This feature bridges the gap between academic model development and practical usability.

However, there are limitations. The current system operates on static data and does not yet support real-time transaction streams. Furthermore, while Random Forest is effective, it lacks interpretability, which may limit its use in regulated financial environments. Future incorporation of explainable AI (XAI) techniques such as SHAP or LIME could help address this challenge.

6. CONCLUSION

This study proposes a machine learning approach for credit card fraud detection using a real-world anonymized dataset, involving data preprocessing, handling class imbalance, model selection, and evaluation with robust metrics, followed by deployment through a user-friendly Streamlit web interface. Among the models tested, the Random Forest classifier performed best, achieving 99.95% accuracy, an F1-score of 84.62%, and an ROC-AUC of 0.983, demonstrating strong ability to differentiate fraudulent from legitimate transactions. The Streamlit app supports both manual and batch detection with real-time feedback, serving as a practical prototype for integration into larger fraud management systems. While the research meets its goals by combining technical rigor and usability, future work includes real-time system integration, secure cloud deployment, deep learning enhancements, and applying explainable AI tools to improve model interpretability, establishing a solid foundation for both academic and real-world applications in financial fraud detection.

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